

INFLATION

A persistent and sustained increase in the general price level over a given period of time.

Persistent increase means that once the price level starts rising it continues to increase. The reason is that an initial increase in the prices will lead to a decrease in the purchasing power or real income of labors who bargain for higher wages. This increase in the wage rate leads to a rise in the cost of production causing profits to fall. In order to maintain the profits the producer increases the prices of their goods. This will again reduce the real income of labors and they bargain for higher wages causing costs to increase again causing a second round of rising prices to begin. This process provides persistency in the price level and is called the wage price spiral. So the persistent increase in the price level is mainly because of labor market adjustments.

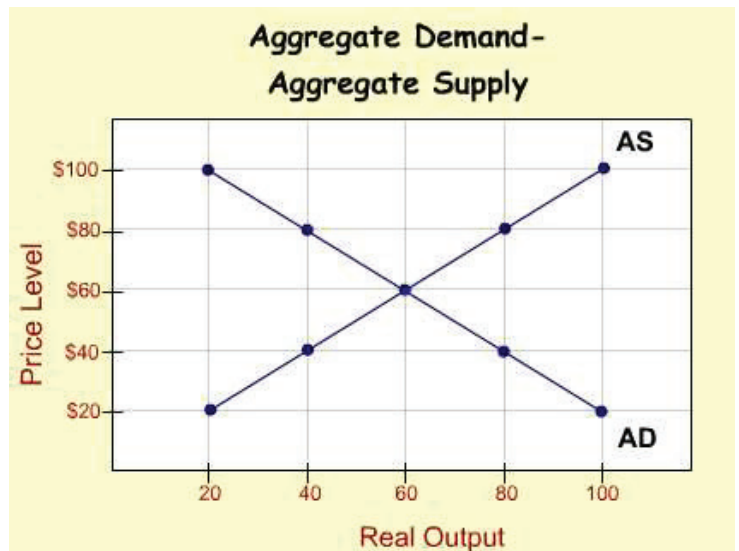
Generalized increase in prices means that the prices of all the goods and services which have been selected to calculate inflation have to be considered i.e. inflation is the average rise in the price s of all these goods and services.

Aggregate demand i.e. total demand of the economy which comprises of consumption, investment, government expenditure and net exports

Net exports is export revenue minus import expenditure - that is balance of trade which can be in surplus when export revenue exceeds import expenditure or it can be deficit when import expenditure exceeds export revenue

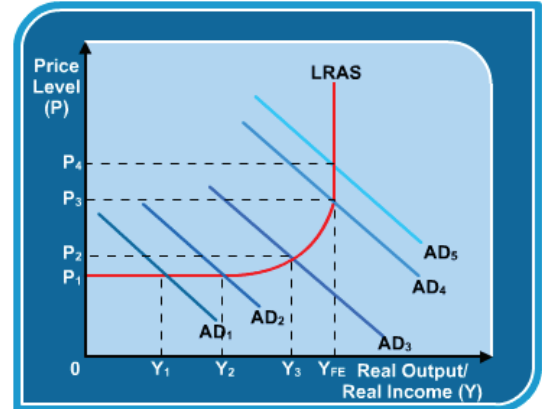
$$AD = C + I + G + (X - M)$$

Aggregate supply is the total supply of the economy.

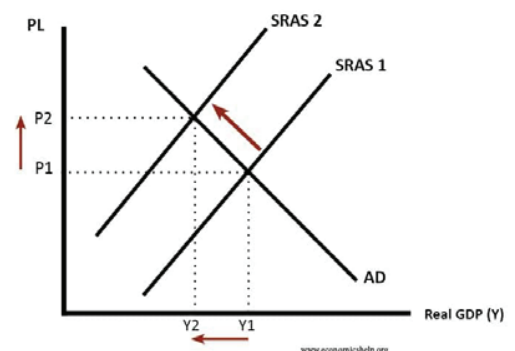


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CAUSES OF INFLATION: There are two broad types of factors which lead to inflation in the economy i.e. demand pull factors and cost push factors. The demand pull factors cause aggregate demand to increase with given aggregate supply causing price level to rise which is called demand pull inflation. On the other hand the cost push factors lead to an increase in the costs of production causing aggregate supply to fall a price level to rise which is called cost push inflation.



1. **Demand pull inflation** will be severe when the economy is operating at or near full employment level of output. There can be different situations where there is full employment of resources and in these situations an increase in aggregate demand will cause severe inflation. Some of these situations are:
 - During and after the war: During the war, most of the resources are engaged in the fighting of war leaving very few resources for the production of consumer goods. This creates severe shortages of these goods causing prices to rise rapidly. After the war most of the resources are engaged in the reconstruction of the economy which again creates shortages of consumer goods which lead to demand pull inflation.
 - Excessive exports: When a country tries to achieve surplus in the balance of payment, it tends to export large quantities of goods and services. This creates shortage of goods and services leading to demand pull inflation.
 - Government expenditure: An important cause of demand pull inflation is expansionary fiscal policy of the government. Especially when an economy is near full employment level of output and when a government adopts expansionary fiscal policy. Its expenditure exceeds the tax revenues i.e. the government budget goes into deficit which is usually financed through borrowing from the central bank. When the government spends this money in the economy it leads to an increase in money supply which leads to an increase in aggregate demand and hence demand pull inflation.
2. **COST PUSH FACTORS:** There are factors which tend to increase the cost of production causing the profits of the producers to reduce. As a result the producers react by increasing the prices of goods and services to shift part or



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whole of the increased cost of production on the consumers.

Cost push inflation also occur when the price of an imported raw material increase which significantly increase the cost of production. This can happen as a result of an increase in the world prices of raw material or because of depreciation of domestic currency which has made the imported raw material more expensive in terms of domestic currency.

Devaluation in the exchange rate is also likely to cause inflation; firstly this will cause an increase in the prices of imports which will be passed on to consumers. Secondly there will be an increase in exports and therefore higher aggregate demand. An increase in the wage rate without increase in the productivities of labor will lead to cost push inflation. E.g. government intervention of minimum wage legislation will increase the cost of producing a given level of national output leading to a fall in aggregate supply. The producers will pass the burden of increased cost on the consumers in the form of increased prices of goods and services.

An increase in indirect tax is also an important cause of cost push inflation. Indirect tax such as value added tax (VAT) is imposed on goods and services which increases the cost of production and producers pass some or entire tax on consumers in the form of increased prices.

3. **Imported inflation:** This occurs in an economy if one country is importing a raw material and the country which is exporting that raw material faces inflation. Imported country will keep on buying that raw material as it is a necessity/inelastic demand i.e. a change in price brings a smaller percentage change in quantity demand. This will increase the cost of the imported country, hence bringing inflation.
4. **Monetary inflation:** This occurs when there is an increase in money supply or fall in interest rate. Fall in interest rate will increase consumption and investment in the economy thus increasing aggregate demand, which in the long run be pushing price level up, resulting in inflation.

An increase in money supply reduces interest rate which influences savers, borrowers and exchange rate. A reduction in interest rate will reduce saving and encourage spending, along with that more will be borrowed thus increase in investment. Importantly a fall in interest rate reduces exchange rate that is depreciation which make exports cheaper and imports to be expensive for our domestic residents, these actions can result in balance of trade surplus. All the above actions will result in rise in aggregate demand which creates shortage and therefore there will be rise in price level.

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EFFECTS OF INFLATION: Inflation affects both the economy of a country and its social conditions, as well as the political and moral lives of its inhabitants. However, the economic effects of inflation are stated and described below:

Inflation does not equally affect the incomes of different groups of people. Creditors and fixed income earners, like pensioners whose incomes are fixed in money terms for a long period, suffer most as the purchasing power of their incomes is substantially curtailed. Creditors will lose in the sense that they are repaid in money which has less purchasing power than that of money lent. Thus all properties and incomes that have fixed values, such as insurance policies, bonds, pensions, rental incomes, and salaries depreciate to extent of inflation. Those who gain from the rising prices or find inflation beneficial to them include businessmen, debtors, who include the government as well, and the shareholders. Businessmen will gain because their costs of production will not increase as soon as the price level rises, so their profit margin will rise. Debtors gain from inflation by having to repay less than what they owed in terms of its real value, since the real value of their debts depreciates. Shareholders whose shares do not carry fixed rate will benefit from large dividends from higher profits of the business firms.

Inflation can adversely affect the **balance of trade** because exports become less competitive in international market causing export demand to fall. If domestic prices are rising at a much faster rate in other countries, the volume of exports is most likely to fall since no country will demand for dearer exports. The amount by which the demand for exports is reduced is determined by the elasticity of demand for exports. If the demand for exports is relatively inelastic, then the rise in prices of exports will have little effect on the demand for exports. If on the other hand, the demand for exports is relatively elastic then the demand for exports will fall more than proportionate increase in their prices. Since the domestic rate inflation is greater than in other countries, imports to increase. The effect of inflation on the volume of imports will again depend on the elasticity of demand for imports. An elastic demand for imports will lead to a more than proportionate increase in the demand for imports. Both these changes in the demand for exports and imports will have an adverse effect on the balance of trade.

In time of inflation the **value of money falls**. Therefore, the people have less incentive to save since the value of savings will be eroded by rising prices. Saving now becomes less attractive because postponement of consumption means having to pay higher prices later on.

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COSTS AND BENEFITS OF LOW INFLATION: There are many benefits of low inflation. Firstly, if inflation is low and stable firms will be more confident and optimistic to invest, this will lead to an increase in aggregate supply and enable higher rates of economic growth in the future. Also, if government is committed to keeping inflation within a certain target this may help inflation expectations are low.

Low rate of inflation keeps the real income or the purchasing power of labors fairly stable and hence the costs of production will also remain stable. Low rate of inflation also helps in achieving a more efficient allocation of resources. When the prices of goods and services increase, the relative profitability of goods and services change and the producers tend to allocate more resources in the production of more profitable goods and services. Secondly, they tend to employ the existing unemployed leading to a more efficient allocation of resources.

High inflation has other costs such as **menu costs**, which is the cost of changing price lists. If inflation in the domestic country is higher than elsewhere, domestic goods will become less competitive causing a fall in exports, and possibly **deterioration in the current account** of the balance of payments.

Inflation is more likely to conflict with unemployment. To keep inflation rate low, the government adopts a contractionary fiscal policy i.e. it reduces the aggregate demand for goods and services, may be by increasing the tax rate. This reduced aggregate demand will make the producers to reduce output causing unemployment to increase.

In **conclusion** low inflation has many benefits which help to improve the economic performance of the economy such as increased investment. However in some circumstances keeping inflation low may be unsuitable for the economy. If there is a fall in aggregate supply and the economy is already in recession, keeping to the inflation target may cause increased unemployment and lower growth which is very undesirable. Therefore the government should perhaps aim for low inflation but have a degree of flexibility if this appears unsuitable to the current economic climate.

Measurement of inflation and deflation

To measure rises and falls in the price level, governments construct price indices (also referred to as price indexes). These show the change in general price level in percentage terms over time. One of the main price indices used is the consumer prices index (CPI).

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Constructing a price index

There are a number of stages in constructing a price index. These include selecting a base year, finding out how households spend their money, attaching weights to items of expenditure, finding out price changes from a range of trade outlets and then constructing a weighted price index.

Selecting a base year

Government statisticians try to select a relatively standard year in which there were no dramatic changes, as a base year. The base year is then given a figure of 100 and the price levels in other years are compared to this figure. For example, if the base year is 2015, it would mean that if the price index in 2018 was 123, the general price level had risen by 23% between 2015 and 2018.

Finding out how households spend their money

In calculating the average rise in prices, it is important to know how people spend their money. This is because a price change in an item on which people spend a large proportion of their total expenditure will have more impact on the cost of living than an item on which they spend a relatively small proportion. If, for example, the price of water rose, it would hit the pockets of most of the population much more than a rise in the price of a trip in an air balloon, something most people will not buy and those who do, will buy infrequently.

To find out the spending patterns of people, government officials carry out surveys of household expenditure. In New Zealand, a sample of approximately 3000 households is used. These households are asked to keep a record of their expenditures. From the information collected, government officials work out the main commodities being bought by the households. This enables them to decide which items to include in the price index and what weights to attach to each of them. If people stop buying a product or their expenditure on it falls to a very small figure, it will be removed from the index. The weights reflect the proportion spent on the items. For example, if on an average, households spend \$120 of their total expenditure of \$600 on food – food will be given a weightage of $\frac{1}{5}$ or 20%. Table 31.1 shows the categories of products in the UK's CPI and their respective weights with, for example, 10.3% of UK households' expenditure going on food and soft drinks.

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	Category	Weight (%)
a	Food and non-alcoholic beverages	10.3
b	Alcohol and tobacco	4.2
c	Clothing and footwear	7.1
d	Housing and household services	12.0
e	Furniture and household goods	5.9
f	Health	2.8
g	Transport	15.3
h	Communication	3.2
i	Recreation and culture	14.8
j	Education	2.5
k	Restaurants and hotels	12.3
l	Miscellaneous goods and services	9.6

Table 31.1: The weights in the UK's CPI in 2016

Source: based on Office for National Statistics Consumer Price Inflation: 2016 Weights <https://www.ons.gov.uk/file3rd March>)

Household spending patterns are reviewed each year with new family expenditure surveys. If these reveal, for example, that people are spending a greater percentage on recreation and culture, and a lower percentage on food and non-alcoholic beverages, the weights of these items in price index will be altered to reflect these changes.

Finding out price changes

Each month government officials find out information about prices. In the UK, about 130 000 price quotations are found for 650 different items. These are obtained from shops, post offices, power companies, train companies and a range of other outlets. From this information, the government estimates the change in prices.

Constructing a weighted price index

Having assigned weights to different items included in the index and measured the change in their prices over time, the final stage is to multiply the weights by the new price index for each category of products and to calculate the change in general price level. For example, consumers may spend \$40 on food, \$10 on housing, \$25 on transport and \$25 on entertainment. This gives a total expenditure of \$100. The price of food may have risen by 10%, the price of housing may have fallen by 5%, the price of transport may not have

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changed and entertainment may have risen in price by 8%. The information would then be used to perform the calculation shown in Table 31.2.

Category	Weight		Price index		Weighted price index
Food	4/10	×	110	=	44.0
Housing	1/10	×	95	=	9.5
Transport	1/4	×	100	=	25.0
Entertainment	1/4	×	108	=	27.0
					105.5

Table 31.2: Weighted price index

The price index has risen by 5.5%. The change in the price level could also have been calculated rather more directly, as shown in Table 31.3.

Category	Weight		Price change (%)		Weighted price change (%)
Food	4/10	×	10	=	4
Housing	1/10	×	-5	=	-0.5
Transport	1/4	×	0	=	0
Entertainment	1/4	×	8	=	2
					5.5

Table 31.3: Weighted price change

Fiscal policy: It involves the use of government spending and tax policies to influence the aggregate demand in order to achieve certain macro-economic objective. Government expenditures are of different nature and are done for various reasons:

1. **To provide public and merit goods:** These are the goods which people would like to have and which many people regard as necessities but these goods are not provided by the private producers or they are underprovided. Example of public goods include defense, streetlights, flood control scheme and so on. Examples of merit goods are health and education services.
2. **Helping those in need:** Government provides certain welfare payments to the citizens who are unable to earn e.g. retired people receive pensions, elderly people receive old age benefits, unemployed receive unemployment benefits and so on.

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3. **Supporting industry and employment:** Large sums of public money are spent on schemes to improve industrial efficiency and to create more jobs.
4. **Managing the economy:** Government expenditure is a significant part of total expenditure in the economy. When private producer allocate insufficient resources to achieve a particular economic growth then the government increases its expenditures on different development projects to create income and employment in the economy, thereby achieving faster rates of economic growth.
5. **Providing subsidies to the producers:** One of the significant part of government expenditure is the provision of subsidies to the producers to help them in reducing the costs.

Another part of fiscal policy is the different types of taxes which it imposes. These taxes may be levied on the incomes of people, on the goods and services and on the wealth of people. These taxes are categorizes as direct and indirect taxes.

Direct taxes: These are taxes which are levied on the incomes an wealth of people e.g. income tax (tax placed on the wages and salaries), corporation taxes (imposed on profits of firms). These taxes are described as direct taxes because the person who is responsible for paying the tax and to bear the burden of the tax.

Indirect taxes: These are the taxes which are levied on the goods and services or the spending in the economy e.g. the sales tax and the value added tax. These taxes are described as indirect taxes because the person responsible for paying the tax may pass on all or part of the burden of the tax in the form of higher prices.

Taxes are imposed in different ways and are categorized as **regressive, progressive and proportional**. A progressive tax is the one where those with higher incomes pay a larger percentage of their incomes e.g. income taxes are progressive. A regressive tax is the one where people with lower income pay a larger percentage of their income as tax e.g. indirect tax such as sales tax act regressively. A proportional tax is the one where all the tax payers pay the same percentage of their income as taxes e.g. corporation and wealth tax.

Governments use **progressive tax system** to achieve a more equitable distribution of income and wealth in the economy. A steeply progressive tax system may discourage the labor i.e. the incentive to work will fall. People will try to rely on unemployment benefits. It may also lead to tax evasion. These factors will reduce the government tax revenue and the aim of redistributing income will not be achieved. A progressive tax system will lead to a fall in consumption level in the economy causing the output to fall

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and unemployment to rise thereby creating bigger difference between the rich and the poor. It may increase inefficiency among the industries because a high corporate tax leaves lesser after tax profits causing investments and output level to fall.

Fiscal policy can be expansionary or contractionary. An expansionary fiscal policy involves rising of government expenditures and lowering of taxes which lead to a budget deficit. This deficit can be financed through borrowings from the central bank which is called public sector borrowing requirement (PSBR). A contractionary fiscal policy involves reducing the government expenditure or raising taxes. When the government tax revenue exceeds the government expenditure the budget will be in surplus and the government will be able to repay the debts that have been accumulated. This is known as public sector debt repayment (PSDR).

Monetary policy: It is the policy of government in which the central bank changes the money supply and the rate of interest to achieve certain macroeconomic objectives such as reducing inflation and unemployment for e.g. if the prices are rising and the government fears an increase in the rate of inflation, it will reduce the money supply or increase the interest rate. This is referred to as contractionary monetary policy. Similarly if the government wants to reduce unemployment, it will stimulate AD either by increasing the money supply or by reducing the interest rate. This is referred to as expansionary monetary policy.

Supply-side policy

Supply-side policy measures are designed to increase aggregate supply and hence increase productive potential. Such policy measures seek to increase the quantity and quality of resources, and raise the efficiency of product and factor markets. These policy measures include improving education and training, cutting direct taxes and benefits, reforming trade unions and privatization.

Improving education and training

Improving education and training is designed to increase the skills of workers. If successful, this would make workers more productive. Higher labour productivity can reduce costs of production and raise the quality of the products produced. Both effects can result in the country's firms being able to sell more and so is likely to encourage them to expand. More skilful workers are also likely to be more occupationally mobile. This should mean that supply would adjust more quickly to changes in demand.

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There is, however, the possibility that the education and training may not be in the right areas. For example, devoting resources to teaching children flower arranging may be less useful than devoting more resources to teaching them mathematics. Training people to work as taxi drivers will also serve little purpose if driverless cars are about to be widely used.

Lowering direct taxes and increasing incentives

The intention behind cutting direct taxes is to increase the incentive to work and invest. Reducing income tax rates will increase the reward from working, especially if combined with a cut in unemployment benefit. Such measures may make the unemployed search for work more actively and may increase the labour force by encouraging more people to seek employment.

Of course, some workers may respond to lower income tax by working fewer hours. This may be because they were happy with their pay and may prefer to have more leisure rather than more pay. In addition, increasing the incentive to work will not be effective if there are not jobs available for the unemployed and those currently inactive to do. Indeed, cutting unemployment benefit and other benefits could make unemployment worse. This is because the unemployed and, for example, the disabled will have less income to spend. This may result in a fall in aggregate demand and so output and employment.

Cutting corporation tax may increase the incentive to invest. This is because it is likely to make firms more willing and able to invest. The greater willingness will come from them being able to keep more of any profits they earn. The greater ability will come from the rise in funds they will have to invest.

If, however, firms lack confidence in the future, they may not invest even with lower corporation tax. If firms do not think they will be able to sell more goods and services in the future, they will not want to expand. A firm when deciding whether to invest takes into account not just the rate of corporation tax. As well as confidence, it also takes into account, for example, advances in technology.

Deregulation

Deregulation is the reduction, or elimination, of rules and regulations that have been enforced by laws. The intentions behind deregulation are to remove barriers to entry to markets and to reduce the costs of complying with the rules and regulations. It will also reduce the cost to the government of regulating the industries and occupations affected.

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If deregulation does increase competition, it may increase efficiency and so lower costs of production and prices. Removing rules and regulations, however, does not guarantee that a monopoly will not develop. There are arguments for regulating monopolies, given their market power. Imposing restrictions on who can undertake particular occupations may also protect consumers. For example, most people would not want the government to remove the requirement that surgeons have to possess medical qualifications. There are, however, debates in some cases about what qualifications are needed to undertake particular occupations.

Privatisation

As with deregulation, privatisation seeks to increase competition and efficiency by increasing the role of market forces. Moving industries into the private sector may put more pressure on them and provide them with a greater incentive to respond to changes in consumer demand, and to provide them with high quality products at low prices. If they do not, they will go out of business, but, if they do, they could make high profits. Privatisation will increase productive capacity, if private sector firms invest more and work more efficiently than state owned enterprises.

Again, as with deregulation, privatisation will not necessarily ensure greater competition. Especially over time, a monopoly may develop. A private sector firm may be less inclined to take into account social costs and social benefits than a state-owned enterprise and may be more willing to make workers redundant.

Labour market reforms

Labour market reforms are designed to make labour markets work more efficiently. The intention is to increase the quality, quantity and flexibility of labour. There is some overlap between labour market reforms and the measures mentioned above. For example, the reforms can include better training and the removal of barriers to entry, such as excessive qualifications, and barriers to exit from labour markets, such as long contracts.

Making it easier for employers to hire and fire workers, is one way that some governments seek to increase the efficiency of labour markets. This is likely to make it easier for firms to adjust their supply to changing market conditions. It may also encourage firms to employ more workers as they will know that they will not have to continue to pay them, should demand for their products fall. There is a risk, however, that firms may not spend as much on training their workers if they think the workers may not be with them for very long.

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Another labour market reform is reform of trade unions. This may make labour more productive. If the power of trade unions is reduced, they may be less inclined to engage in industrial action. There may be less disruption to production and trade union members may be more willing to be flexible in what tasks they do and the hours they work. If trade unions have been intimidating their members into taking industrial action, there may be a case for reducing their power. In some cases, though, reducing trade union power may reduce the benefits that trade unions may provide not only for their members, but also for employers and the economy as a whole. It is also possible that trade union reforms may give too much power to employers which could lower the pay of workers and increase their working hours. Being paid less and working more hours may reduce the motivation of workers and make them tired which would be likely to reduce their productivity.

Subsidies

A government may provide subsidies to the firms in particular industries for a number of reasons connected to increasing the performance of markets. For example, new small firms might be subsidized in order to increase competition in markets. Firms may also be subsidized to encourage them to buy new capital equipment. Of course, consideration would have to be given as to whether this is the best use of government spending and there is the risk that the firms may become dependent on the subsidies.

Effects of supply-side policy on government macroeconomic aims

In the long run, all the government's macroeconomic aims have the potential to benefit from supply-side policy. Increasing aggregate supply enables an economy to continue to grow in a non-inflationary way. Figure 28.1 shows aggregate supply rising in line with aggregate demand. Such a combination enables output and employment to increase without inflation.

Increasing productive potential and efficiency can improve an economy's balance of payments position. Producing better quality, and cheaper, products can increase exports and reduce imports.

In the case of some supply-side policy measures there is a time lag involved. For example, it can take some time before the effects of improved education and privatisation are experienced. In addition, some supply-side measures can be expensive.

Increasing the effectiveness of macroeconomic policies

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Besides using supply-side policy measures in the long run to improve macroeconomic performance, there are a number of other ways through which a government can try to ensure that it achieves all its macroeconomic aims.

One is by using a number of policies. A Nobel Prize winning economist, Jan Tinbergen, suggested that a government needs to use one policy measure for each of its objectives. So, for example, if a government wants to stimulate economic growth and reduce imports, it may provide investment grants to firms and place a tax on imports.

Another way to try to ensure that all of its aims are achieved, is to have as much, and as accurate information, as possible. One vital piece of information is the size of the multiplier effect of any increase in aggregate demand. If aggregate demand is raised too much, for example, it will push up the price level.

Governments also try to decide and implement their policies relatively quickly. If there is a delay in introducing policies, there is a danger that economic activity undergoes a change and the policy measures may actually harm the economy. For example, a period of high unemployment may lead the government to cut income tax, and to raise aggregate demand and employment. If, however, by the time the measure is introduced, aggregate demand is increasing anyway, it may increase inflationary pressure.